

W02 | VMware 網路模式與雙 VM 排錯

網路配置

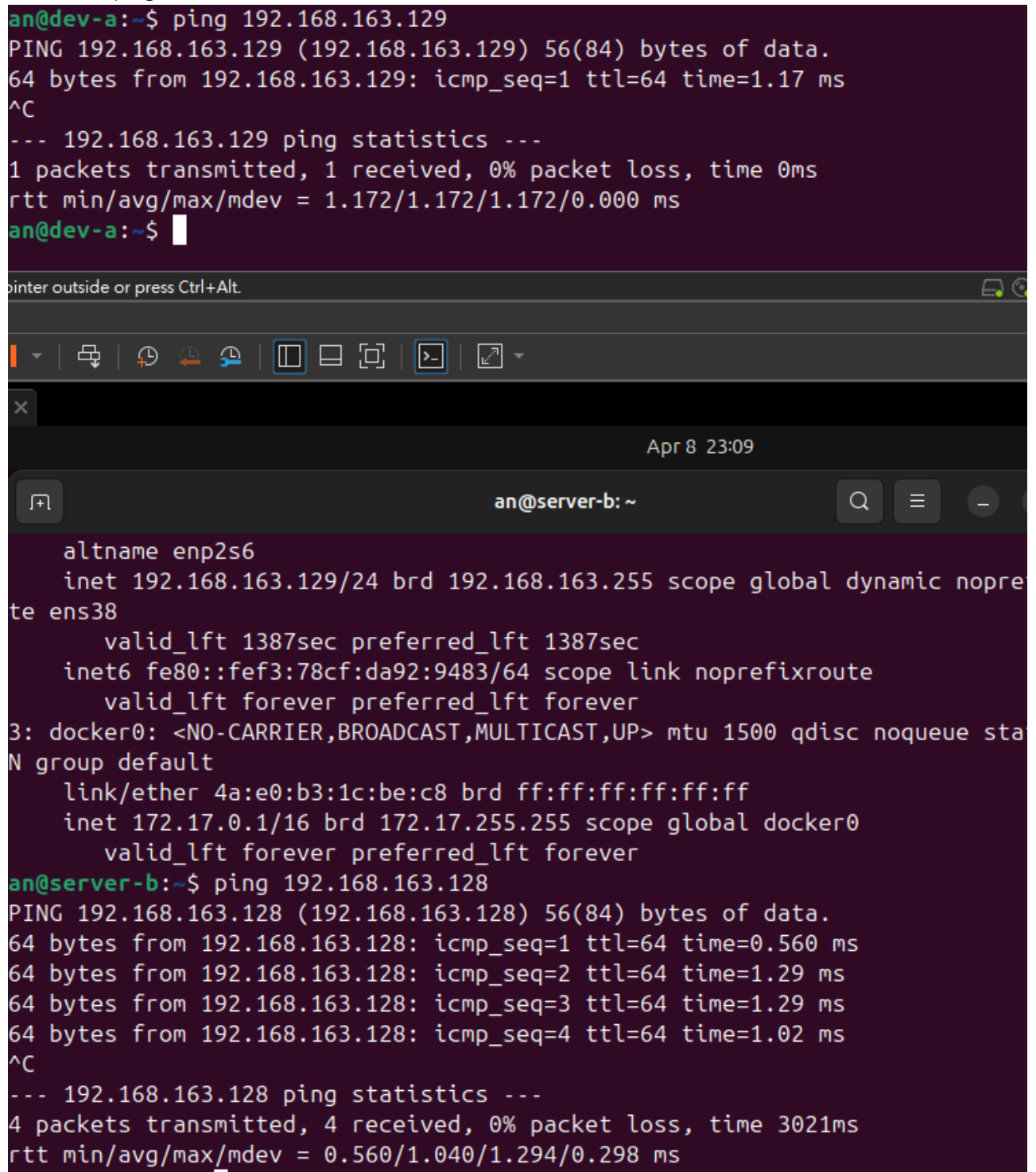
VM	網卡	模式	IP	用途
dev-a	NIC 1	NAT	192.168.192.129	上網
dev-a	NIC 2	Host-only	192.168.163.128	內網互連
server-b	NIC 1	Host-only	192.168.163.129	內網互連

連線驗證紀錄

- ☑ dev-a NAT 可上網：ping google.com 輸出

```
an@dev-a:~$ ping google.com
PING google.com (142.250.66.78) 56(84) bytes of data.
64 bytes from google.com (142.250.66.78): icmp_seq=1 ttl=128 time=12.0 ms
64 bytes from google.com (142.250.66.78): icmp_seq=2 ttl=128 time=11.8 ms
64 bytes from google.com (142.250.66.78): icmp_seq=3 ttl=128 time=12.1 ms
64 bytes from google.com (142.250.66.78): icmp_seq=4 ttl=128 time=12.5 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 11.828/12.126/12.545/0.260 ms
```

-  雙向互 ping 成功：貼上雙方 ping 輸出



The screenshot shows two terminal windows. The top window, titled 'an@dev-a:~', shows a successful ping to 192.168.163.129. The bottom window, titled 'an@server-b: ~', shows the output of the 'ip netns exec' command for the 'ens38' interface, followed by a successful ping to 192.168.163.128.

```
an@dev-a:~$ ping 192.168.163.129
PING 192.168.163.129 (192.168.163.129) 56(84) bytes of data.
64 bytes from 192.168.163.129: icmp_seq=1 ttl=64 time=1.17 ms
^C
--- 192.168.163.129 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.172/1.172/1.172/0.000 ms
an@dev-a:~$
```

```
an@server-b: ~$ ip netns exec ens38
altname enp2s6
inet 192.168.163.129/24 brd 192.168.163.255 scope global dynamic nopre
te ens38
    valid_lft 1387sec preferred_lft 1387sec
inet6 fe80::fef3:78cf:da92:9483/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue sta
N group default
    link/ether 4a:e0:b3:1c:be:c8 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
an@server-b:~$ ping 192.168.163.128
PING 192.168.163.128 (192.168.163.128) 56(84) bytes of data.
64 bytes from 192.168.163.128: icmp_seq=1 ttl=64 time=0.560 ms
64 bytes from 192.168.163.128: icmp_seq=2 ttl=64 time=1.29 ms
64 bytes from 192.168.163.128: icmp_seq=3 ttl=64 time=1.29 ms
64 bytes from 192.168.163.128: icmp_seq=4 ttl=64 time=1.02 ms
^C
--- 192.168.163.128 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3021ms
rtt min/avg/max/mdev = 0.560/1.040/1.294/0.298 ms
```

-  SSH 連線成功：ssh <user>@<ip> "hostname" 輸出

```
an@dev-a:~$ ssh an@192.168.163.129
The authenticity of host '192.168.163.129 (192.168.163.129)' can't be established.
ED25519 key fingerprint is SHA256:ONCBfATm6XUAqNn6VUoItAtOEbByOJGANLgTvVeaRE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.163.129' (ED25519) to the list of known hosts.
an@192.168.163.129's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-19-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

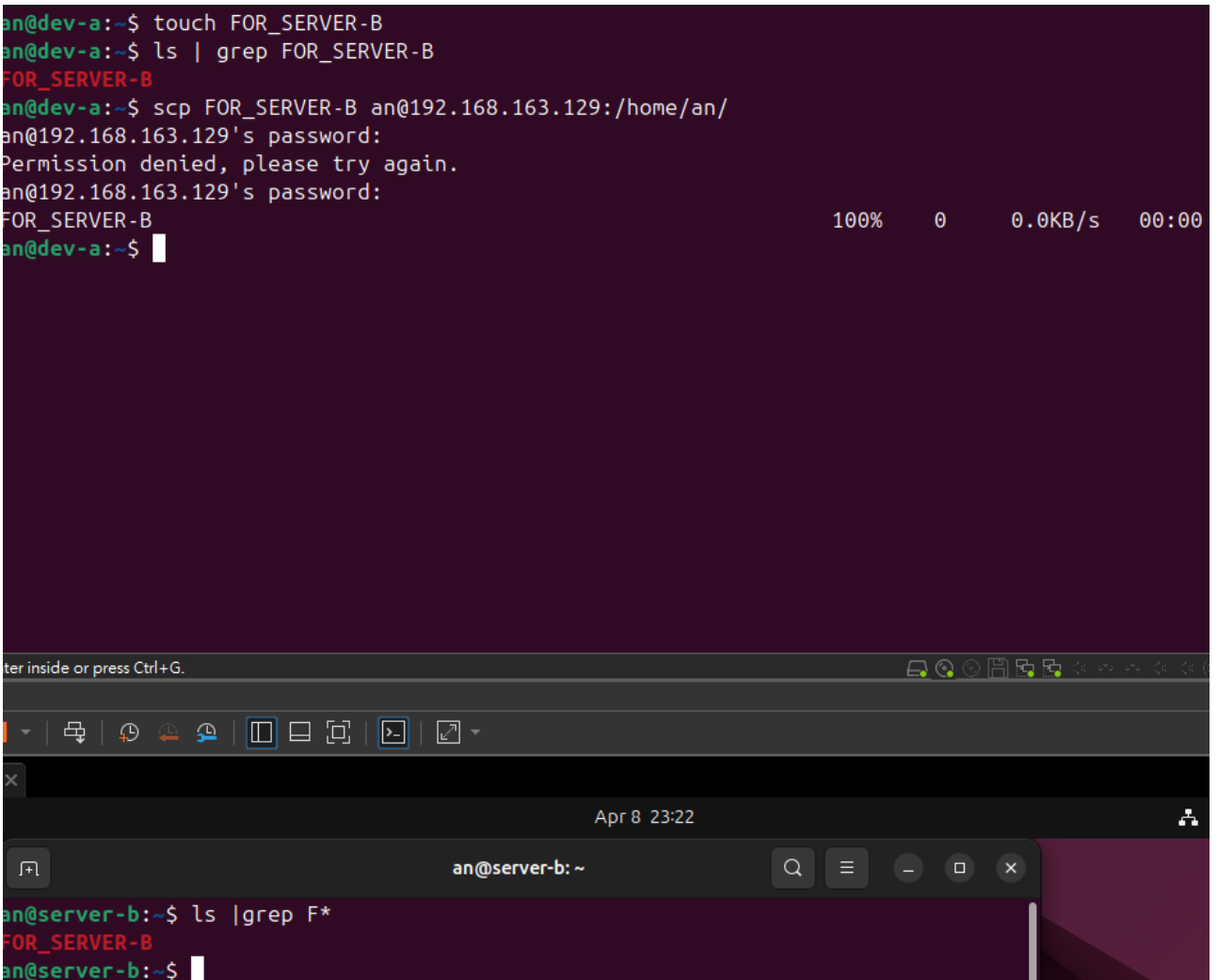
10 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

an@server-b:~$
```

-  SCP 傳檔成功：cat /tmp/test-from-dev.txt 在 server-b 上的輸出

```
an@dev-a:~$ touch FOR_SERVER-B
an@dev-a:~$ ls | grep FOR_SERVER-B
FOR_SERVER-B
an@dev-a:~$ scp FOR_SERVER-B an@192.168.163.129:/home/an/
an@192.168.163.129's password:
Permission denied, please try again.
an@192.168.163.129's password:
FOR_SERVER-B                                100%   0    0.0KB/s   00:00
an@dev-a:~$
```



The screenshot shows a terminal window with a dark purple background. The top part shows the command sequence on 'dev-a' to create a file 'FOR_SERVER-B' and attempt to SCP it to 'server-b'. The SCP command fails with a 'Permission denied' error. The bottom part shows a new terminal window titled 'an@server-b: ~' where the command 'ls | grep F*' is executed, resulting in the output 'FOR_SERVER-B'.

- server-b 不能上網：ping 8.8.8.8 失敗輸出

```
an@server-b:~$ ping 8.8.8.8
ping: connect: Network is unreachable
an@server-b:~$
```

故障演練一：介面停用

項目	故障前	故障中	回復後
server-b 介面狀態	UP	DOWN	UP
dev-a ping server-b	成功	失敗	成功
dev-a SSH server-b	成功	失敗	成功

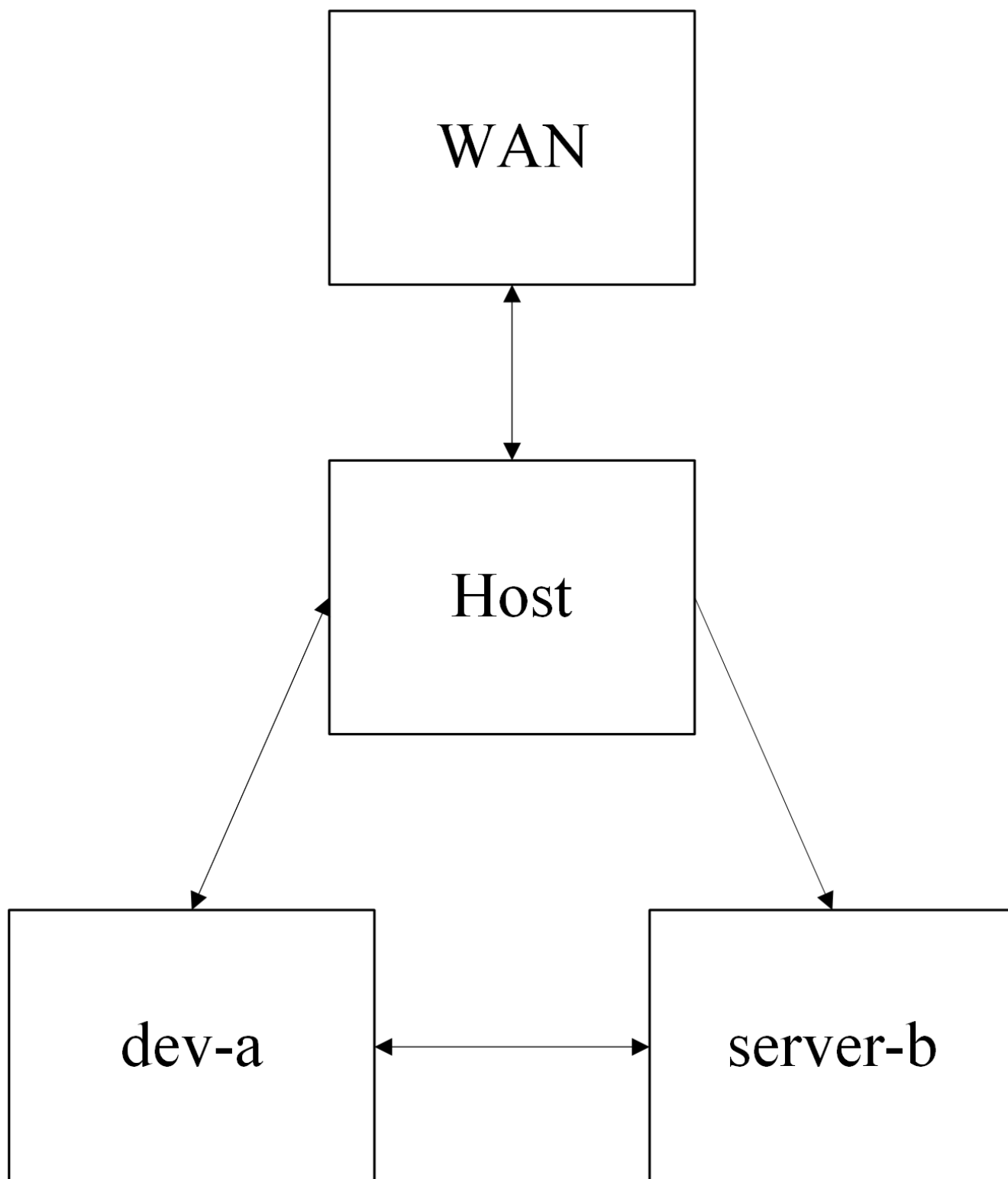
故障演練二：SSH 服務停止

項目	故障前	故障中	回復後
ss -tlnp grep :22	有監聽	無監聽	有監聽
dev-a ping server-b	成功	成功	成功
dev-a SSH server-b	成功	Connection refused	成功

排錯順序

- L2 (資料連結層) 排錯：**確認網卡狀態與實體連線。
 - 使用命令：ip link show 檢查介面是否為 UP，或於 VMware 中檢查網卡是否為「已連線」。
- L3 (網路層) 排錯：**確認 IP 配置與路由連通性。
 - 使用命令：ip a 檢查 IP 設定，ping <目標 IP> 測試點對點連通性，以及 ip route 查看路由表。
- L4 (傳輸層) 排錯：**確認服務 Port 是否正常監聽，以及是否被防火牆阻擋。
 - 使用命令：在目標機使用 ss -tlnp | grep :22 檢查服務狀態，或從來源機使用 nc -zv <目標 IP> 22 測試埠口是否開放。

網路拓樸圖



排錯紀錄

- 症狀：從 **dev-a** 無法透過 SSH 連線至 **server-b**，錯誤訊息顯示為 **Connection refused**。
- 診斷：首先執行 **ping 192.168.163.129** (L3 排錯)，發現回應正常，排除網路斷線問題。接著在 **server-b** 執行 **ss -tlnp | grep :22** (L4 排錯)，發現無任何輸出，確認 SSH 服務未啟動。
- 修正：在 **server-b** 執行 **sudo systemctl start ssh** (或 **sshd**) 來重新啟動 SSH 服務。
- 驗證：於 **server-b** 再次執行 **ss -tlnp | grep :22** 確認已正常監聽 22 Port；回到 **dev-a** 執行 **ssh an@192.168.163.129 "hostname"** 成功回傳主機名稱。

設計決策

為什麼 **server-b** 只配置 **Host-only** 模式而不給 **NAT**？基於「最小權限原則 (Principle of Least Privilege)」與網路安全隔離設計。**server-b** 在架構中定位為內部伺服器，不應直接暴露於網際網路，也不需主動向外網連線。將其限制在 Host-only 網段能有效縮小攻擊面 (Attack Surface)。若未來 **server-b** 需連外 (如：下載更新)，應讓具有雙網卡 (NAT + Host-only) 的 **dev-a** 兼任 Proxy 或 NAT Router 來集中控管對外流量，以實現更安全的網路架構分層。